INCH-POUND
MIL-M-38510/101H
30 October 2001
SUPERSEDING
MIL-M-38510/101G
08 MAY 1987

MILITARY SPECIFICATION

MICROCIRCUITS, LINEAR, OPERATIONAL AMPLIFIER, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

Inactive for new design after 13 July 1995

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, operational amplifiers. Two product assurance classes and a choice of case outlines and lead finish are provided for each type and are reflected in the complete part number.

1.2 Part number. The complete part number shall be in accordance with MIL-M-38510.

1.2.1 Device types. The device types shall be as shown in the following:

Device type	Circuit
01	Single operational amplifier - internally compensated
02	Dual operational amplifier - internally compensated
03	Single operational amplifier - externally compensated
04	Single operational amplifier - externally compensated
05	Dual operational amplifier - externally compensated 1/
06	Dual operational amplifier - externally compensated 1/
07	Single operational amplifier, high speed
08	Dual operational amplifier - internally compensated

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1/ Device types 05 and 06 may be monolithic, or they may consists of two separate, independent die.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAS, 3990 East Broad St., Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.2.3 Case outlines. The case outlines shall be designated as follows:

Outline letter	Descriptive designator	Terminals	Package style
A 2/	GDFP5-F14 or CDFP6-F14	14	Flat pack
B 2/	GDFP4-14	14	Flat pack
C	GDIP1-T14 or CDIP2-T14	14	Dual-in-line
D	GDFP1-F14 or CDFP2-F14	14	Flat pack
E	GDIP1-T16 or CDIP2-T16	16	Dual-in-line
F	GDFP2-F16 or CDFP3-F16	16	Flat pack
G	MACY1-X8	8	Can
H	GDFP1-F10 or CDFP2-F10	10	Flat pack
I	MACY1-X10	8	Can
P	GDIP1-T8 or CDIP2-T8	8	Dual-in-line
Z	GDFP1-G10	10	Flat pack with gullwing leads
2	CQCC1-N20	20	Square leadless chip carrier

1.3 Absolute maximum ratings.

Supply voltage range (V_{CC}) ..... ±22 V dc 3/
Input voltage range ..... ±20 V dc 4/
Differential input voltage range ..... ±30 V dc 5/
Input current range ..... -0.1 mA to +10 mA
Storage temperature range ..... -65°C to +150°C
Output short-circuit duration ..... Unlimited 6/
Lead temperature (soldering, 60 seconds) ..... +300°C
Junction temperature (T_J) ..... +175°C 7/

1.4 Recommended operating conditions.

Supply voltage (V_{CC}) ..... ±5 V dc to 20 V dc
Ambient temperature range (T_A) ..... -55°C to +125°C

2/ Inactive package case outline.
3/ Voltages in excess of these may be applied for short-term tests if voltage difference does not exceed 44 volts.
4/ For supply voltages less than ±20 V dc, the absolute maximum input voltage is equal to the supply voltage.
5/ For device types 04, 06, and 07 only, this rating is ±1.0 V unless resistances of 2 kΩ or greater are inserted in series with the inputs to limit current in the input shunt diodes to the maximum allowable value.
6/ Short circuit may be to ground or either supply. Rating applies to +125°C case temperature or +75°C ambient temperature.
7/ For short term test (in the specific burn-in and life test configuration when required and up to 168 hours maximum) T_J = +275°C.

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1.5 Power and thermal characteristics.

Case outlines	Maximum allowable power dissipation	Maximum θ_{JC}	Maximum θ_{JA}
A,B,D	350 mW at T_A = +125°C	60°C/W	140°C/W
C,E,P	400 mW at T_A = +125°C	35°C/W	120°C/W
G	330 mW at T_A = +125°C	40°C/W	150°C/W
I	350 mW at T_A = +125°C	40°C/W	140°C/W
H	330 mW at T_A = +125°C	60°C/W	150°C/W
F	400 mW at T_A = +125°C	35°C/W	120°C/W
Z	330 mW at T_A = +125°C	21°C/W	225°C/W still air
			142°C/W 500 LFPM
2	8/ at T_A = +125°C	60°C/W	120°C/W

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Method Standard for Microelectronics.
MIL-STD-1835 - Interface Standard For Electronic Component Outlines.

(Unless otherwise indicated, copies of the specifications, standards, and handbooks are available from the Standardization Documents Order Desk, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein the text of this document shall takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

8/ P_D = 102 mW for device type 01. P_D = 75 mW for device type 03. P_D = 149 mW for device type 04.

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3. REQUIREMENTS

3.1 Qualification. Item furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.3 and 6.2).

3.2 Item requirements. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein. The individual item requirements for devices marked with the "Q" certification mark shall be in accordance with MIL-PRF-38535 and as specified herein or as modified in the device manufacturer's Quality Management (QM) plan. The modification in the QM plan shall not affect the form, fit, or function as described herein.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and 1.2.3 herein.

3.3.1 Circuit diagram and terminal connections. The circuit diagram and terminal connections shall be as specified on figure 1.

3.3.2 Schematic circuits. The schematic circuits shall be submitted to the preparing activity prior to inclusion of manufacturer's device in this specification and shall be submitted to the qualifying activity and preparing activity (DSCC-VAS) as a prerequisite for qualification. All qualified manufacturer's schematics shall be maintained by the preparing activity and will be available upon request. A typical schematic circuit shall be as specified on figure 2.

3.3.3 Case outlines. The case outlines shall be as specified in 1.2.3.

3.4 Lead material and finish. Lead material and finish shall be in accordance with MIL-M-38510.

3.5 Electrical performance characteristics. The following electrical performance characteristics apply over the full operating ambient temperature range of -55°C to +125°C and for supply voltages ±5 V dc to ±20 V dc, unless otherwise specified (see table I).

3.5.1 Offset null circuits. Each amplifier having nulling inputs (device types 01, 02, 03, 05, and 07) shall be capable of being nulled 1 mV beyond the specified offset voltage limits for $-55°C \leq T_A \leq +125°C$ using the circuits of figure 3.

3.5.2 Frequency compensation. Device types 01, 02, 07, and 08 shall be free of oscillation when operated in a unity gain non-inverting mode with no external compensation and a source resistance of ≤ 10 kΩ, and when operated in any test condition specified herein. Device types 03, 04, 05, and 06 shall be free from oscillation when compensated with a 30 pF capacitor for all gain configurations or a 3 pF capacitor when used with a gain of 10.

3.6 Rebonding. Rebonding shall be in accordance with MIL-M-38510.

3.7 Electrical test requirements. Electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.8 Marking. Marking shall be in accordance with MIL-M-38510.

3.9 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 49 (see MIL-M-38510, appendix E).

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TABLE I. Electrical performance characteristics. 1/

Test	Symbol	Conditions $-55°C \le T_A \le +125°C$ see figure 4 unless otherwise specified	Group A subgroups	Device type	Limits		Unit
					Min	Max	
Input offset voltage	V_{IO}	$R_S = 50\ \Omega$ 2/	1	01,02, 08	-3	+3	mV
				03,05	-2	+2	
				04,06	-0.5	+0.5	
				07	-4	+4	
			2,3	01,02, 08	-4	+4	
				03,05	-3	+3	
				04,06	-1	+1	
				07	-6	+6	
Input offset voltage temperature sensitivity	$\Delta V_{IO} / \Delta T$		2	01,02, 08	-15	+15	µV/°C
				03,05	-18	+18	
				04,06	-5	+5	
				07	-50	+50	
			3	01,02, 03,05, 08	-15	+15	
				04,06	-5	+5	
				07	-50	+50	

See footnotes at end of table.

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TABLE I. Electrical performance characteristics – Continued. 1/

Test	Symbol	Conditions $-55°C \le T_A \le +125°C$ see figure 4 unless otherwise specified	Group A subgroups	Device type	Limits		Unit
					Min	Max	
Input offset current	I_{IO}	2/	1	01,02, 08	-30	+30	nA
				03,05	-10	+10	
				04,06	-0.2	+0.2	
				07	-40	+40	
			2,3	01,02, 08	-70	+70	
				03,05	-20	+20	
				04,06	-0.4	+0.4	
				07	-80	+80	
Input offset current temperature sensitivity	$\Delta I_{IO} / \Delta T$		2	01,02, 08	-500	+500	pA/°C
				03,05	-200	+200	
				04,06	-2.5	+2.5	
				07	-1000	+1000	
			3	01,02, 08	-200	+200	
				03,05	-100	+100	
				04,06	-2.5	+2.5	
				07	-1000	+1000	

See footnotes at end of table.

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TABLE I. Electrical performance characteristics – Continued. 1/

Test	Symbol	Conditions $-55°C \le T_A \le +125°C$ see figure 4 unless otherwise specified	Group A subgroups	Device type	Limits		Unit
					Min	Max	
Input bias current	$+I_{IB}$	2/	1,2	01,02, 08	-0.1	+110	nA
				03,05	-0.1	+75	
				07	-0.1	+250	
			1	04,06	-0.1	+2.0	
			2		-1.0	+2.0	
			3	01,02, 08	-0.1	+265	
				03,05	-0.1	+100	
				04,06	-0.1	+3.0	
				07	-0.1	+400	
	$-I_{IB}$		1,2	01,02, 08	-0.1	+110	
				03,05	-0.1	+75	
				07	-0.1	+250	
			1	04,06	-0.1	+2.0	
			2		-1.0	+2.0	
			3	01,02, 08	-0.1	+265	
				03,05	-0.1	+100	
				04,06	-0.1	+3.0	
				07	-0.1	+400	

See footnotes at end of table.

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TABLE I. Electrical performance characteristics – Continued. 1/

Test	Symbol	Conditions $-55°C \le T_A \le +125°C$ see figure 4 unless otherwise specified	Group A subgroups	Device type	Limits		Unit
					Min	Max	
Power supply rejection ratio	+PSRR	$+V_{CC}$ = 10 V, R_S = 50 Ω, $-V_{CC}$ = -20 V	1	01,02, 03,05, 08	-50	+50	µV/V
				04,06	-16	+16	
				07	-100	+100	
			2,3	01,02, 03,05, 08	-100	+100	
				04,06	-16	+16	
				07	-150	+150	
	-PSRR	$+V_{CC}$ = 20 V, R_S = 50 Ω, $-V_{CC}$ = -10 V	1	01,02, 03,05, 08	-50	+50	
				04,06	-16	+16	
				07	-100	+100	
			2,3	01,02, 03,05, 08	-100	+100	
				04,06	-16	+16	
				07	-150	+150	
Input voltage common mode rejection	CMR	$\pm V_{CC}$ = 20 V, V_{IN} = ±15 V, R_S = 50 Ω	1,2,3	01,02, 03,05, 07,08	80		dB
				04,06	96		

See footnotes at end of table.

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TABLE I. Electrical performance characteristics – Continued. 1/

Test	Symbol	Conditions $-55°C \le T_A \le +125°C$ see figure 4 unless otherwise specified	Group A subgroups	Device type	Limits		Unit
					Min	Max	
Adjustment for input offset voltage 3/	V_{IO} ADJ(+)	$\pm V_{CC}$ = 20 V	1,2,3	01,02, 08	+5		mV
				03,05	+4		
				04,06	No external ADJ		
				07	+7		
Adjustment for input offset voltage 3/	V_{IO} ADJ(-)	$\pm V_{CC}$ = 20 V	1,2,3	01,02, 08		-5	mV
				03,05		-4	
				04,06		No extern al ADJ	
				07		-7	
Output short-circuit current (for positive output)	$I_{OS}(+)$	V_{CC} = 15 V, t ≤ 25 ms 4/	1,2,3	01,02, 03,05, 08	-60		mA
				04,06	-15		
				07	-65		
Output short-circuit current (for negative output)	$I_{OS}(-)$	$\pm V_{CC}$ = 15 V, t ≤ 25 ms 4/	1,2,3	01,02, 03,05, 08		+60	mA
				04,06		+15	
			1,2	07		+65	
			3			+80	

See footnotes at end of table.

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TABLE I. Electrical performance characteristics – Continued. 1/

Test	Symbol	Conditions $-55°C \le T_A \le +125°C$ see figure 4 unless otherwise specified	Group A subgroups	Device type	Limits		Unit
					Min	Max	
Supply current	I_{CC}	$\pm V_{CC} = \pm 15$ V 5/	1	01,02, 08		+3.8	mA
				03,05		+3	
				04,06		+0.6	
				07		+8	
			2	01,02, 08		+3.4	
				03,05		+2.5	
				04,06		+0.6	
				07		+7	
			3	01,02, 08		+4.2	
				03,05		+3.5	
				04,06		+0.8	
				07		+9	
Output voltage swing (maximum)	V_{OP}	$\pm V_{CC} = 20$ V, $R_L = 10$ kΩ	4,5,6	01-06, 08	±16		V
				07	±17		
		$\pm V_{CC} = 20$ V, $R_L = 2$ kΩ		01,02, 03,05, 08	±15		
				04,06	Not speci-fied		
				07	±16		

See footnotes at end of table.

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TABLE I. Electrical performance characteristics – Continued. 1/

Test	Symbol	Conditions $-55°C \le T_A \le +125°C$ see figure 4 unless otherwise specified	Group A subgroups	Device type	Limits		Unit
					Min	Max	
Open loop voltage gain 6/ (single ended)	$A_{VS}(\pm)$	$\pm V_{CC}$ = 20 V, 7/ R_L = 2 kΩ, 10 kΩ V_{OUT} = ±15 V	4	01,02, 03,05, 07,08	50		V/mV
				04,06	80		
			5,6	01,02, 03,05, 08	25		
				04,06	40		
				07	32		
Open loop voltage gain 6/ (single ended)	A_{VS}	$\pm V_{CC}$ = 5 V, 7/ R_L = 2 kΩ, 10 kΩ V_{OUT} = ±2 V	4,5,6	01,02, 03,05, 07,08	10		V/mV
				04,06	20		
Transient response rise time	$TR_{(tr)}$	See figure 5 8/	7,8A,8B	01,02, 03,05, 08		+800	ns
				04,06		+1000	
				07		+40	
Transient response overshoot	$TR_{(OS)}$	See figure 5 8/	7,8A,8B	01,02, 03,05, 08		+25	%
				04,06, 07		+50	
Slew rate 9/	SR(+)	V_{IN} = ±5 V, A_V = 1, see figure 5	7,8B	01,02, 08	+0.3		V/µs
				03,05	10/		
				04,06	+0.05		
				07	+40		

See footnotes at end of table.

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TABLE I. Electrical performance characteristics – Continued. 1/

Test	Symbol	Conditions $-55°C \le T_A \le +125°C$ see figure 4 unless otherwise specified	Group A subgroups	Device type	Limits		Unit
					Min	Max	
Slew rate 9/	SR(+)	$V_{IN} = \pm 5$ V, $A_V = 1$, see figure 5	8A	01,02, 03,05, 08	+0.3		V/µs
				04,06	+0.05		
				07	+30		
Slew rate 9/	SR(-)	$V_{IN} = \pm 5$ V, $A_V = 1$, see figure 5	7,8B	01,02, 08	+0.3		V/µs
				03,05	10/		
				04,06	+0.05		
				07	+40		
			8A	01,02, 03,05, 08	+0.3		
				04,06	+0.05		
				07	+30		
Settling time 11/	$t_S(+)$	See figure 5	12	07		800	ns
			13A,13B			1200	
	$t_S(-)$		12			800	
			13A,13B			1200	
Channel separation	CS	$\pm V_{CC} = \pm 20$ V, see figure 6, $T_A = +25°C$	7	02,05, 06,08	80		dB

See footnotes at end of table.

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TABLE I. Electrical performance characteristics – Continued. 1/

Test	Symbol	Conditions $-55°C \le T_A \le +125°C$ see figure 4 unless otherwise specified	Group A subgroups	Device type	Limits		Unit
					Min	Max	
Noise (referred to input) broadband	NI(BB)	$\pm V_{CC}$ = 20 V, T_A = +25°C, bandwidth = 5 kHz	7	01-06, 08		15	µVrms
				07		25	
Noise (referred to input) popcorn	NI(PC)	$\pm V_{CC}$ = 20 V, T_A = +25°C, bandwidth = 5 kHz	7	01,02, 04,06, 08		40	µVpk
				03,05, 07		80	

1/ For devices marked the “Q” certification mark, the parameters listed herein maybe guaranteed if not tested to the limits specified herein in accordance with the manufacturer’s QM plan.

2/ Tests at common mode V_{CM} = 0 V, V_{CM} = -15 V, and V_{CM} = +15 V.

3/ $V_{IO(ADJ)}$ is not performed on device type 02, case I only, or on device type 08 for either case G or P.

4/ Continuous short circuit limits will be considerably less than the indicated test limits. Continuous I_{OS} at $T_A \le +75°C$ will cause T_J to exceed the maximum of +175°C. For dual devices, I_{OS} is measured one channel at a time.

5/ Value shown is for single devices (01, 03, 04) only. For dual devices (02, 05, 06, and 08) this limit is for single devices.

6/ Note that gain is not specified at $V_{IO(ADJ)}$ extremes. Some gain reduction is usually seen at $V_{IO(ADJ)}$ extremes. For closed loop applications (closed loop gain less than 1,000), the open loop tests (AVS) prescribed herein should guaranteed a positive, reasonably linear, transfer characteristic. They do not, however, guarantee that the open loop gain is linear, or even positive, over the operating range. If either of these requirements exist (positive open loop gain or open loop gain linearity), they should be specified in the individual procurement document as additional requirements.

7/ R_L = 10 kΩ only for device types 04 and 06.

8/ For transient response tests, C_F = 10 pF for device types 01, 02, 03, 04, 05, 06, and 08. Device type 07, C_F = 47 pF. C_F includes the effects of stray capacitance.

9/ Minimum limit for device 08 is 0.4 V/µs at all temperatures.

10/ Minimum limits for device types 03 and 05 are 0.2 V/µs at -55°C and 0.3 V/µs at both +25°C and +125°C.

11/ Settling time is waived for method 5004, MIL-STD-883 except for device type 07.

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TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (method 5004)	1	1
Final electrical test parameters 1/ (method 5004)	1,2,3,4	1,2,3,4
Group A test requirements (method 5005)	1,2,3,4,5,6, 7,8A,8B,12, 13A,13B	1,2,3,4,5,6,7
Group C end point and group B, class S, electrical parameters (method 5005)	1,2,3, and table IV delta limits	1 and table IV delta limits
Additional electrical subgroups For group C periodic inspections	Not applicable	8A,8B,12, 13A,13B
Group D end point electrical parameters (method 5005)	1,2,3	1

1/ PDA applies to subgroup 1 (see 4.2d).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5004, 5005, and 5007, as applicable, of MIL-STD-883, except as modified herein. For devices marked with the "Q" certification mark, sampling and inspection procedures shall be in accordance with MIL-PRF-38535 or as modified in the device manufacturer's Quality Management (QM) plan. The modification in the QM plan shall not effect the form, fit, or function as function as described herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in (method 1015 of MIL-STD-883).

1. For class S devices: Test condition D using the circuit shown on figure 7.

2. For class B devices: Test condition D using the circuit shown on figure 7 or test condition C using the circuit shown on figure 8, or test condition F using the circuit (or equivalent) shown on figure 9.

b. Reverse bias burn-in (method 1015 of MIL-STD-883). For class S devices only using the circuit shown on figure 8.

c. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

d. The percent defective allowable (PDA) for class S and class B devices shall be as specified in MIL-M-38510, based on failures from group A, subgroup 1 test after cooldown as final electrical test in accordance with method 5004 of MIL-STD-883 and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroup 1 after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

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Device type	01			
Case outlines	A,B,C,D	G and P	H	2
Terminal number	Terminal symbol			
1	NC	OFFSET NULL	NC	NC
2	NC	-INPUT	OFFSET NULL	OFFSET NULL (-)
3	OFFSET NULL	+INPUT	-INPUT	NC
4	-INPUT	$-V_{CC}$	+INPUT	NC
5	+INPUT	OFFSET NULL	$-V_{CC}$	-INPUT
6	$-V_{CC}$	OUTPUT	OFFSET NULL	NC
7	NC	$+V_{CC}$	OUTPUT	+INPUT
8	NC	NC	$+V_{CC}$	NC
9	OFFSET NULL	---	NC	NC
10	OUTPUT	---	NC	$-V_{CC}$
11	$+V_{CC}$	---	---	NC
12	NC	---	---	OFFSET NULL (+)
13	NC	---	---	NC
14	NC	---	---	NC
15	---	---	---	OUTPUT
16	---	---	---	NC
17	---	---	---	$+V_{CC}$
18	---	---	---	NC
19	---	---	---	NC
20	---	---	---	NC

NC = No connection

FIGURE 1. Terminal connections.

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Device type	02		03		
Case outlines	A,B,C,D	I	C	G and P	H
Terminal number	Terminal symbol				
1	-INPUT A	OUTPUT A	NC	OFFSET NULL / COMP	NC
2	+INPUT A	$+V_{CC}$ (A) SEE NOTE 2	NC	-INPUT	OFFSET NULL / COMP
3	OFFSET NULL A	-INPUT A	OFFSET NULL / COMP	+INPUT	-INPUT
4	$-V_{CC}$	+INPUT A	-INPUT	$-V_{CC}$	+INPUT
5	OFFSET NULL B	$-V_{CC}$	+INPUT	OFFSET NULL	$-V_{CC}$
6	+INPUT B	+INPUT B	$-V_{CC}$	OUTPUT	OFFSET NULL
7	-INPUT B	-INPUT B	NC	$+V_{CC}$	OUTPUT
8	OFFSET NULL B	$+V_{CC}$ (B) SEE NOTE 2	NC	COMP	$+V_{CC}$
9	$+V_{CC}$ (B) SEE NOTE 2	OUTPUT B	OFFSET NULL	---	COMP
10	OUTPUT B	NC	OUTPUT	---	NC
11	NC	---	$+V_{CC}$	---	---
12	OUTPUT A	---	COMP	---	---
13	$+V_{CC}$ (A) SEE NOTE 2	---	NC	---	---
14	OFFSET NULL A	---	NC	---	---
15	---	---	---	---	---
16	---	---	---	---	---
17	---	---	---	---	---
18	---	---	---	---	---
19	---	---	---	---	---
20	---	---	---	---	---

NC = No connection

FIGURE 1. Terminal connections – Continued.

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Device type	03	04			
Case outlines	2	C	G and P	H	2
Terminal number	Terminal symbol				
1	NC	NC	INPUT COMP	NC	NC
2	OFFSET NULL (-)	INPUT COMP	-INPUT	GUARD	INPUT COMP
3	NC	GUARD	+INPUT	-INPUT	NC
4	NC	-INPUT	$-V_{CC}$	+INPUT	NC
5	-INPUT	+INPUT	NC	GUARD	-INPUT
6	NC	GUARD	OUTPUT	$-V_{CC}$	NC
7	+INPUT	$-V_{CC}$	$+V_{CC}$	OUTPUT	+INPUT
8	NC	NC	OUTPUT COMP	$+V_{CC}$	NC
9	NC	NC	---	OUTPUT COMP	NC
10	$-V_{CC}$	OUTPUT	---	INPUT COMP	$-V_{CC}$
11	NC	$+V_{CC}$	---	---	NC
12	OFFSET NULL (+)	OUTPUT COMP	---	---	NC
13	NC	NC	---	---	NC
14	NC	NC	---	---	NC
15	OUTPUT	---	---	---	OUTPUT
16	NC	---	---	---	NC
17	$+V_{CC}$	---	---	---	$+V_{CC}$
18	NC	---	---	---	NC
19	NC	---	---	---	NC
20	FREQ COMP	---	---	---	OUTPUT COMP

NC = No connection

FIGURE 1. Terminal connections – Continued.

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Device type	05	06	07	
Case outlines	E and F	E and F	C	G and P
Terminal number	Terminal symbol			
1	$+V_{CC}$ (A) SEE NOTE 5	$+V_{CC}$ (A) SEE NOTE 5	NC	COMP A / OFFSET NULL
2	COMP A	OUTPUT COMP A	NC	-INPUT
3	OFFSET NULL / COMP	INPUT COMP A	COMP A / OFFSET NULL	+INPUT
4	-INPUT A	-INPUT A	-INPUT	$-V_{CC}$
5	+INPUT A	+INPUT A	+INPUT	COMP B / OFFSET NULL
6	$-V_{CC}$	$-V_{CC}$	$-V_{CC}$	OUTPUT
7	OFFSET NULL B	NC	NC	$+V_{CC}$
8	OUTPUT B	OUTPUT B	NC	COMP C
9	$+V_{CC}$ (B) SEE NOTE 5	$+V_{CC}$ (B) SEE NOTE 5	COMP B / OFFSET NULL	---
10	COMP B	OUTPUT COMP B	OUTPUT	---
11	OFFSET NULL / COMP B	INPUT COMP B	$+V_{CC}$	---
12	-INPUT B	-INPUT B	COMP C	---
13	+INPUT B	+INPUT B	NC	---
14	OFFSET NULL A	NC	NC	---
15	NC	NC	---	---
16	OUTPUT A	OUTPUT A	---	---

NC = No connection

FIGURE 1. Terminal connections – Continued.

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Device types	07	08
Case outlines	H	G and P
Terminal number	Terminal symbol	
1	NC	OUTPUT A
2	COMP A / OFFSET NULL	-INPUT A
3	-INPUT	+INPUT A
4	+INPUT	$-V_{CC}$
5	$-V_{CC}$	+INPUT B
6	COMP B / OFFSET NULL	-INPUT B
7	OUTPUT	OUTPUT B
8	$+V_{CC}$	$+V_{CC}$
9	COMP C	---
10	NC	---

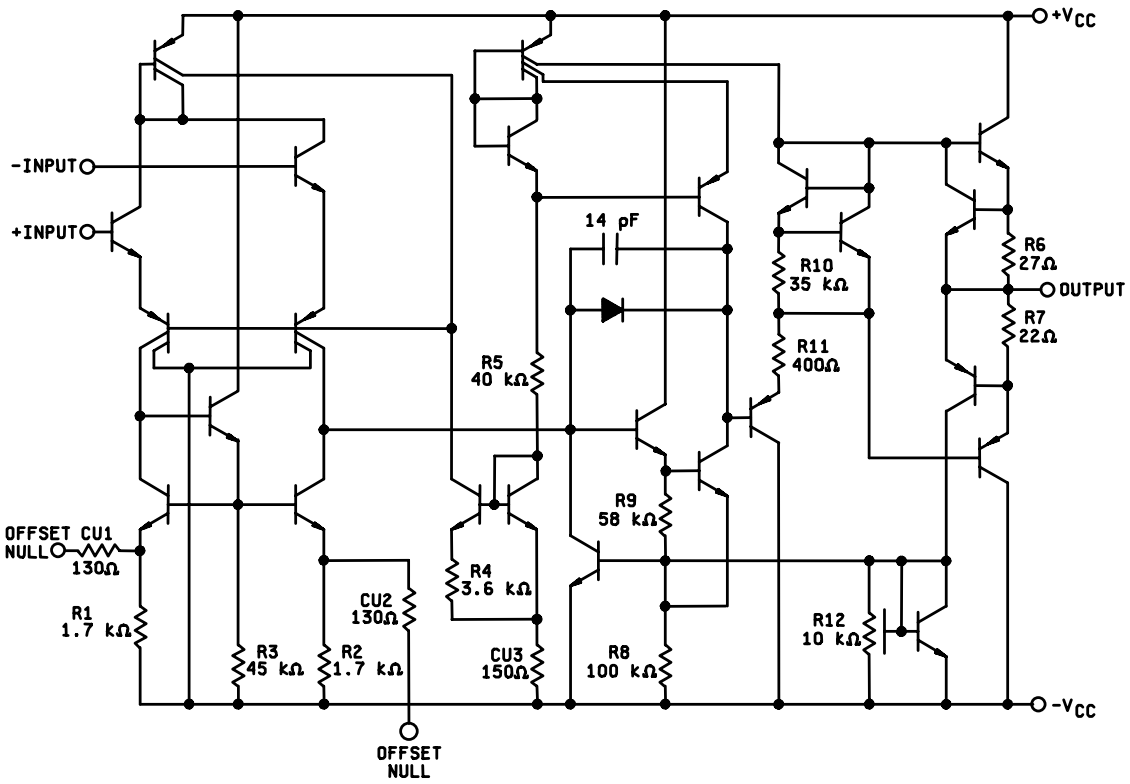
NC = No connection

NOTES:

1. $-V_{CC}$ shall be connected to case of metal packages.
2. For device type 02 only, $+V_{CC}$ (A) and $+V_{CC}$ (B) shall be internally connected.
3. +Input is non-inverting input.
4. -Input is inverting input.
5. For device types 05 and 06 only, $+V_{CC}$ (A) and $+V_{CC}$ (B) shall not be internally connected. (External connection to the same supply voltage recommended).

FIGURE 1. Terminal connections – Continued.

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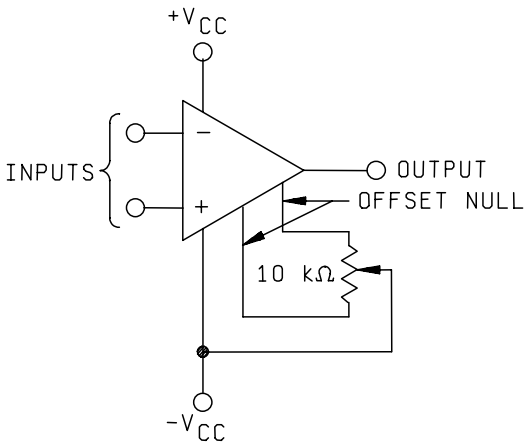


NOTE: Figure shown represents a single channel.
C.U. indicates isolated cross-under.

FIGURE 2. Schematic circuit.

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DEVICE TYPES 01 AND 02



DEVICE TYPES 03 AND 05

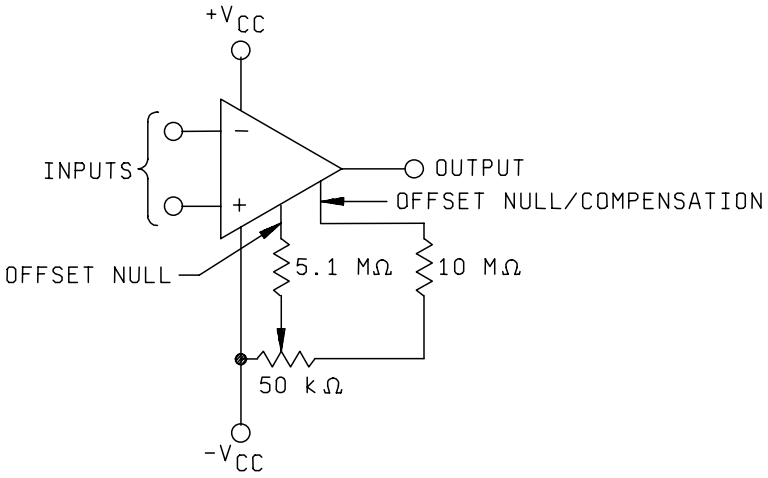


FIGURE 3. Offset null circuits.

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Device type 07

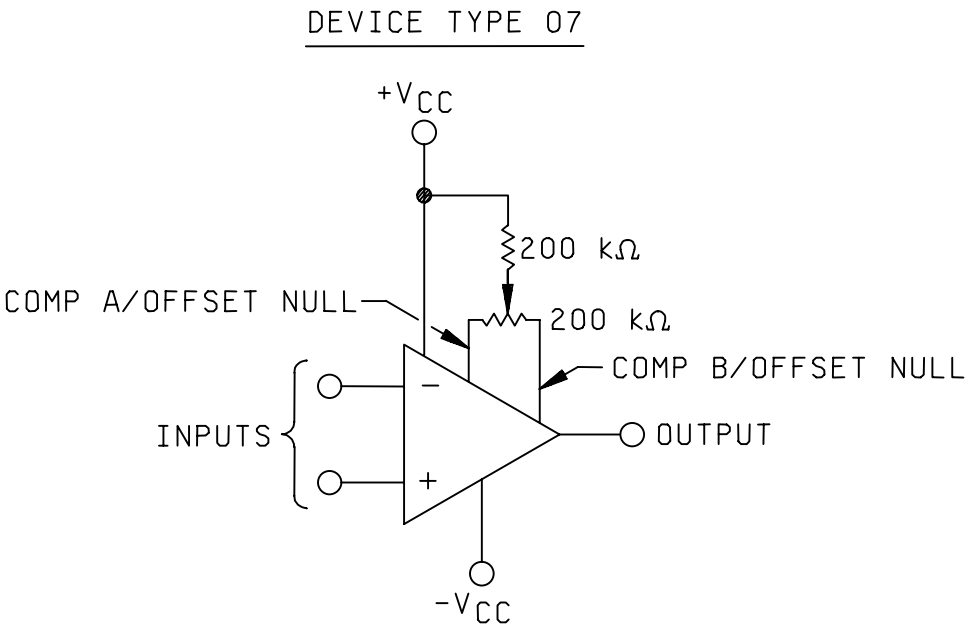


FIGURE 3. Offset null circuits - Continued.

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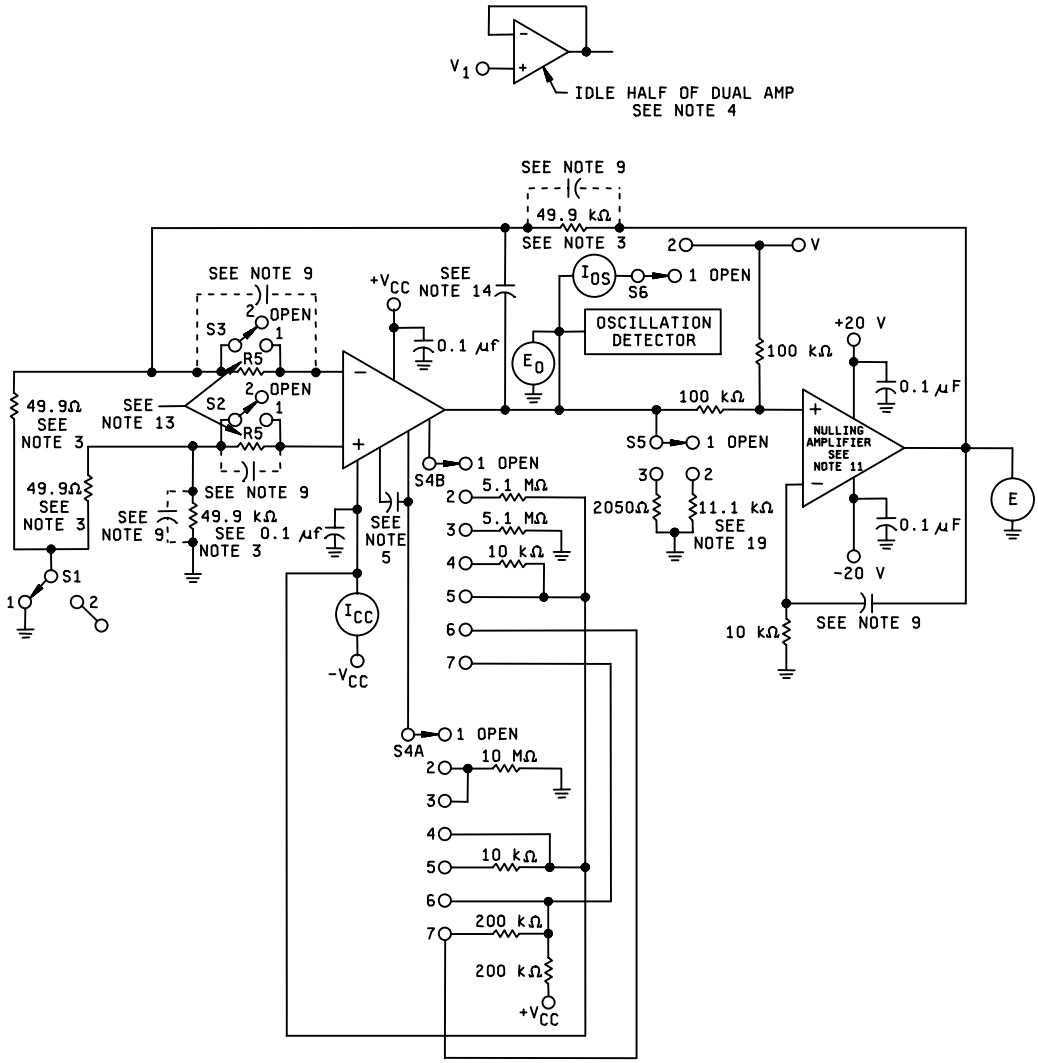


FIGURE 4. Test circuit for static and dynamic tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:16Z
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Parameter	Apply (in volts)			Switch position						Measure		Measured parameter 20/	Units
	+VCC	-VCC	V	S_1	S_2	S_3	S_4	S_5	S_6	Value	Units	Equation	
V_{IO}	35 5 20 5	-5 -35 -20 -5	-15 15 0 0	1 1 1 1	1 1 1 1	1 1 1 1	1 1 1 1	1 1 1 1	1 1 1 1	E1 E2 E3 E4	V	V_{IO} = E1, E2, E3, E4	mV
I_{IO}	35 5 20 5	-5 -35 -20 -5	-15 15 0 0	1 1 1 1	2 2 2 2	2 2 2 2	1 1 1 1	1 1 1 1	1 1 1 1	E5 E6 E7 E8	V	$I_{IO} = ((E1 - E5) \times 10^6) / RS, ((E2 - E6) \times 10^6) / RS,$ 13/ $((E3 - E7) \times 10^6) / RS, ((E4 - E8) \times 10^6) / RS$	nA
$+I_{IB}$	35 5 20 5	-5 -35 -20 -5	-15 15 0 0	1 1 1 1	2 2 2 2	1 1 1 1	1 1 1 1	1 1 1 1	1 1 1 1	E9 E10 E11 E12	V	$+I_{IB} = ((E1 - E9) \times 10^6) / RS, ((E2 - E10) \times 10^6) / RS,$ 13/ $((E3 - E11) \times 10^6) / RS, ((E4 - E12) \times 10^6) / RS$	nA
$-I_{IB}$	35 5 20 5	-5 -35 -20 -5	-15 15 0 0	1 1 1 1	1 1 1 1	2 2 2 2	1 1 1 1	1 1 1 1	1 1 1 1	E13 E14 E15 E16	V	$-I_{IB} = ((E13 - E1) \times 10^6) / RS, ((E14 - E2) \times 10^6) / RS,$ 13/ $((E15 - E31) \times 10^6) / RS, ((E16 - E4) \times 10^6) / RS$	nA
+PSRR	10	-20	5	1	1	1	1	1	1	E18 1/	V	$+PSRR = (E3 - E18) \times 10^2$	µV/V
-PSRR	20	-10	-5	1	1	1	1	1	1	E19 1/	V	$-PSRR = (E3 - E19) \times 10^2$	µV/V
CMR	35 5	-5 -35	-15 +15	1 1	1 1	1 1	1 1	1 1	1 1	E1 1/ E2 1/	V V	$CMR = 20 \log \lvert(30 \times 10^3) / (E1 - E2)\rvert$	dB
V_{IO} 7/ ADJ(+)	20	-20	0	1	1	1	2	1	1	E20	V	V_{IO} ADJ (+) = (E3 - E20)	mV
V_{IO} 7/ ADJ(-)	20	-20	0	1	1	1	3	1	1	E21	V	V_{IO} ADJ (-) = (E3 - E21)	mV
V_{IO} 6/ ADJ(+)	20	-20	0	1	1	1	4	1	1	E22	V	V_{IO} ADJ (+) = (E3 - E22)	mV
V_{IO} 6/ ADJ(-)	20	-20	0	1	1	1	5	1	1	E23	V	V_{IO} ADJ (-) = (E3 - E23)	mV
V_{IO} 15/ ADJ(+)	20	-20	0	1	1	1	6	1	1	E34	V	V_{IO} ADJ (+) = (E3 - E34)	mV
V_{IO} 15/ ADJ(-)	20	-20	0	1	1	1	7	1	1	E35	V	V_{IO} ADJ (-) = (E3 - E35)	mV

FIGURE 4. Test circuit for static and dynamic tests- Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:16Z
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Parameter	Apply (in volts)			Switch position						Measure		Measured parameter 20/	Units
	+VCC	-VCC	V	S1	S2	S3	S4	S5	S6	Value	Units	Equation	
$+I_{OS}$ (output)	15	-15	-15	1	1	1	1	1	2	I_{OS1}	mA	$+I_{OS} = I_{OS1}$	mA
$-I_{OS}$ (output)	15	-15	+15	1	1	1	1	1	2	I_{OS2}	mA	$-I_{OS} = I_{OS2}$	mA
I_{CC}	15	-15	0	1	1	1	1	1	1	I_{CC}	mA	$I_{CC} = I_{CC}$	mA
$+V_{OP}$ R_L = 10 kΩ	20	-20	-20	1	1	1	1	2	1	(E0)1	V	$+V_{OP}$ = (E0)1	V
$-V_{OP}$ R_L = 10 kΩ	20	-20	+20	1	1	1	1	2	1	(E0)2	V	$-V_{OP}$ = (E0)2	V
$+V_{OP}$ R_L = 2 kΩ	20	-20	-20	1	1	1	1	3	1	(E0)3	V	$+V_{OP}$ = (E0)3	V
$-V_{OP}$ R_L = 2 kΩ	20	-20	+20	1	1	1	1	3	1	(E0)4	V	$-V_{OP}$ = (E0)4	V
$+A_{VS}$ 16/ R_L = 2 kΩ	20	-20	-15	1	1	1	1	3	1	E24	V	$+A_{VS}$ = 15 / (E3 - E24)	V/mV
$-A_{VS}$ 16/ R_L = 2 kΩ	20	-20	+15	1	1	1	1	3	1	E25	V	$-A_{VS}$ = 15 / (E25 - E3)	V/mV
A_{VS} 16/ R_L = 2 kΩ	5 5	-5 -5	-2 +2	1 1	1 1	1 1	1 1	3 3	1 1	E26 E27	V	A_{VS} = 4 / (E27 - E26)	V/mV
$+A_{VS}$ 16/ R_L = 10 kΩ	20	-20	-15	1	1	1	1	2	1	E30	V	$+A_{VS}$ = 15 / (E3 - E30)	V/mV
$-A_{VS}$ 16/ R_L = 10 kΩ	20	-20	+15	1	1	1	1	2	1	E31	V	$-A_{VS}$ = 15 / (E31 - E3)	V/mV
A_{VS} 16/ R_L = 10 kΩ	5 5	-5 -5	-2 +2	1 1	1 1	1 1	1 1	2 2	1 1	E32 E33	V	A_{VS} = 4 / (E33 - E32)	V/mV
CMR 3/	20 20	-20 -20	+15 -15	2 2	1 1	1 1	1 1	1 1	1 1	E28 1/ E29 1/	V	CMR = 20 log $\left[(30 \times 10^{3}) / (E28 - E29)\right]$	

FIGURE 4. Test circuit for static and dynamic tests- Continued.

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NOTES:

1/ These voltages in mV shall be measured to four place accuracy to provide required resolution in PSRR and CMR.

2/ Precautions shall be taken to prevent damage to the device under test during insertion into socket and change of switch positions (example, disable voltage supplies, current limit $\pm V_{CC}$, etc.).

3/ If this alternate CMR test is used, these resistors shall be of .01 percent tolerances matched to .001 percent.

4/ Device types 02, 05, and 06 only, test both halves for all tests. The idle half of the dual amplifiers shall be maintained in this configuration where V1 is midway between $+V_{CC}$ and $-V_{CC}$, or the manufacturer has the option to connect the idle half in a V_{IO} configuration such that the inputs are maintained at the same common mode voltage as the device under test.

5/ Compensation: for device types 03, 04, 05, and 06 only, equals 30 pF; for device type 07 only, equals 330 pF (optional).

6/ Device types 01, all case types, and device type 02, case outlines A, B, C, and D only.

7/ Device types 03 and 05 only.

8/ See figure 10. Noise test circuit.

9/ As required, if needed to prevent oscillation. Also, proper wiring procedures shall be followed to prevent oscillation. Loop response and settling time shall be consistent with the test rate such that any value has settled for at least five loop time constants before the value is measured.

10/ Adequate settling time shall be allowed such that each parameter has settled to within five percent of its final value.

11/ The nulling amplifier is an M38510/10101XXX. Saturation of the nulling amplifier is not allowed on test where the "E" value is measured.

12/ All resistors 0.1 percent tolerance except as noted (note 3).

13/ For device types 01, 02, 07, and 08: R_S = 20 kΩ. For device types 03 and 05: R_S = 100 kΩ. For device types 04 and 06: R_S = 5.0 MΩ.

14/ Device type 07 only, this capacitor = 1,000 pF maximum to prevent oscillations.

15/ Device type 07 only.

16/ To minimize thermal drift, the reference voltages for gain measurements (E3 and E4) shall be taken immediately prior to or after the reading corresponding to device gain (E24, E25, E26, E27, E30, E31, E32, and E33). The gain at R_L = 10 kΩ is essentially the gain at R_L = 2 kΩ is influenced by thermal gradients on the die resulting from power dissipation in the output stage. Hence, it is not linear and may not even be a true approximation of the gain between other than the specified operation points.

17/ Any oscillation greater that 300 mV in amplitude (pk - pk) shall be cause for device failure.

18/ Although switches are depicted as toggle switches, any switching mechanism may be used provided the switching action is achieved without adversely affecting the measurement.

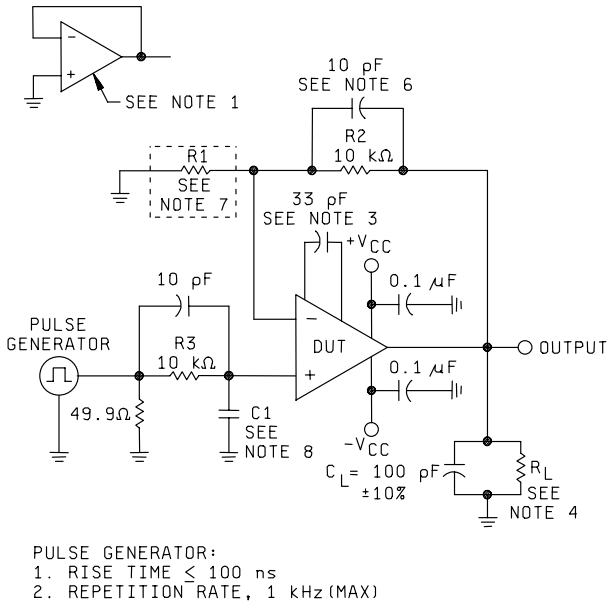
19/ The load resistors (2,050 Ω and 11.1 kΩ) yield effective load resistances of 2 kΩ and 10 kΩ, respectively.

20/ The equations take into account both the gain of 1,000 and the scale factor multiplier, so that the calculated value is in table III units. Therefore, use measured value / units in the equations, example E1 (volts).

FIGURE 4. Test circuit for static and dynamic tests- Continued.

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DEVICE TYPES 01,02,03,04,05,06, AND 08



PULSE GENERATOR:
1. RISE TIME ≤ 100 ns
2. REPETITION RATE, 1 kHz(MAX)

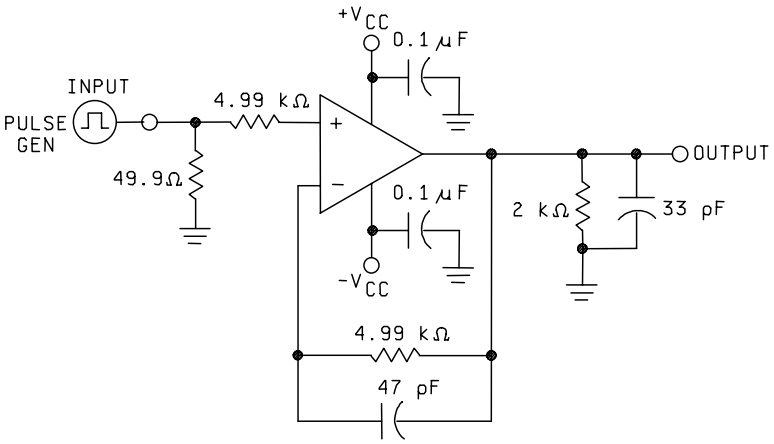
See notes on page 30

Parameter	Pulse generator	Measure	Equation	Units
Rise time (tr) A_V = 1	+50 mV amplitude	t (µs), see waveform 1	$t_r = t$	µs
Overshoot (OS) A_V = 1	+50 mV amplitude	ΔV (mV), see waveform 1	OS = (ΔV / 50) x 100	%
Bandwidth (BW) A_V = 1	+50 mV amplitude	Calculate	BW = 0.35 / tr (µs)	MHz
Slew rate (+SR) A_V = 1	-5 V to +5 V step	ΔV_O (volts), Δt (µs) see waveform 2	$+SR = \lvert\Delta V_O(+) / \Delta t(+)\rvert$	V/µs
Slew rate (-SR) A_V = 1	+5 V to -5 V step	ΔV_O (volts), Δt (µs) see waveform 3	$-SR = \lvert\Delta V_O(-) / \Delta t(-)\rvert$	V/µs

FIGURE 5. Transient response test circuit.

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DEVICE TYPE 07



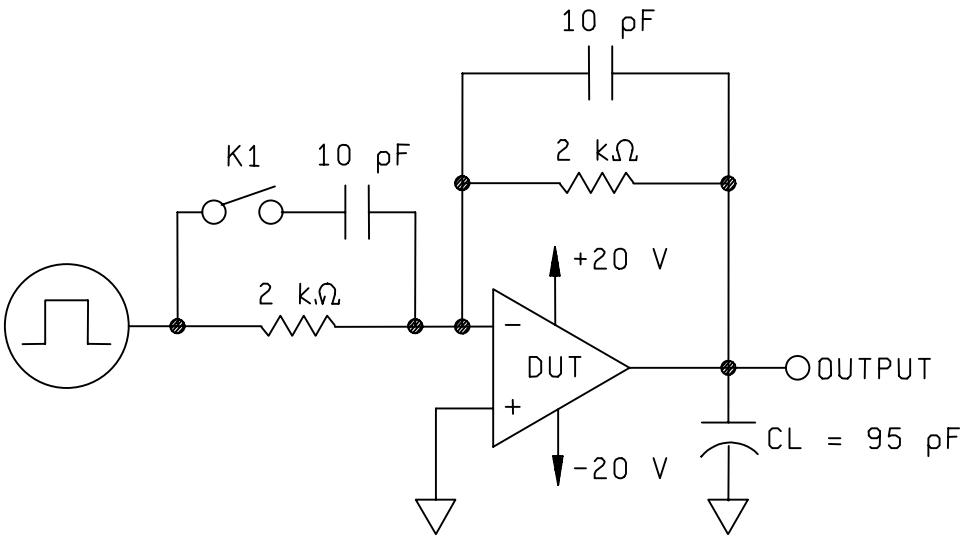
PULSE GENERATOR:
1. RISE TIME ≤ 10 ns
2. REPETITION RATE, 1 kHz (MAX)

See notes on page 30

Parameter	Pulse generator	Measure	Equation	Units
Rise time (tr)	+50 mV amplitude	t (ns), see waveform 1	$t_r = t$	ns
Overshoot (OS)	+50 mV amplitude	ΔV (mV), see waveform 1	OS = (ΔV / 50) x 100	%
Bandwidth (BW)	+50 mV amplitude	Calculate	BW = (0.35×10^3) / tr (ns)	MHz
Slew rate (+SR)	-5 V to +5 V step	ΔV_O (+) (volts), Δt (+) (ns) see waveform 2	+SR = $\lvert \Delta V_O(+) / \Delta t(+) \times 10^{-3} \rvert$	V/µs
Slew rate (-SR)	+5 V to -5 V step	ΔV_O (-) (volts), Δt (-) (ns) see waveform 3	-SR = $\lvert \Delta V_O(-) / \Delta t(-) \times 10^{-3} \rvert$	V/µs
Settling time $t_S(+)$ 5/	-5 V to +5 V step	$t_S(+)$, see waveform 2	$t_S(+) = t_S(+)$	ns
Settling time $t_S(-)$ 5/	+5 V to -5 V step	$t_S(-)$, see waveform 3	$t_S(-) = t_S(-)$	ns

FIGURE 5. Transient response test circuit - Continued.

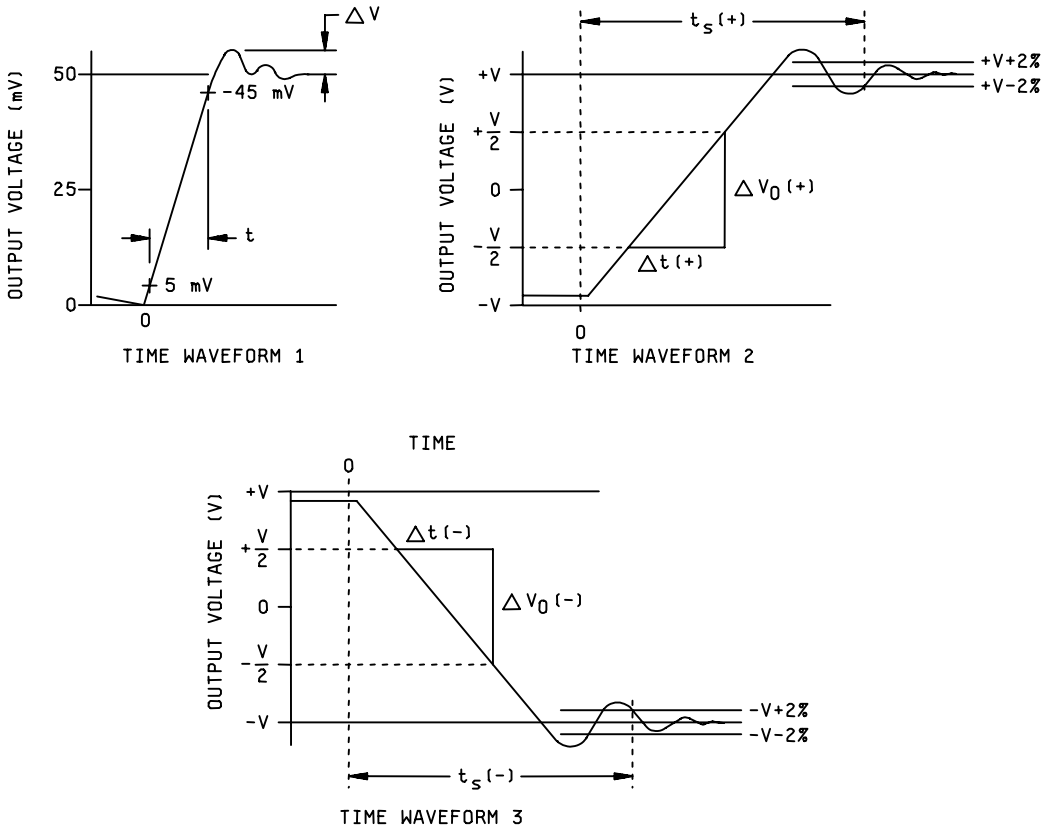
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:16Z
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NOTES:
1. K1 is closed for small tests (Tr and P.O.) and is open for large signal tests (±slew rate, ±Ts).
2. Input signal is a -50 mV to 0 mV pulse train for small signal tests and -5 V to +5 V pulse train for large signal tests.
3. Tr of the input signal is < 10 ns for the small signal tests.

FIGURE 5. Transient response test circuit - Continued.

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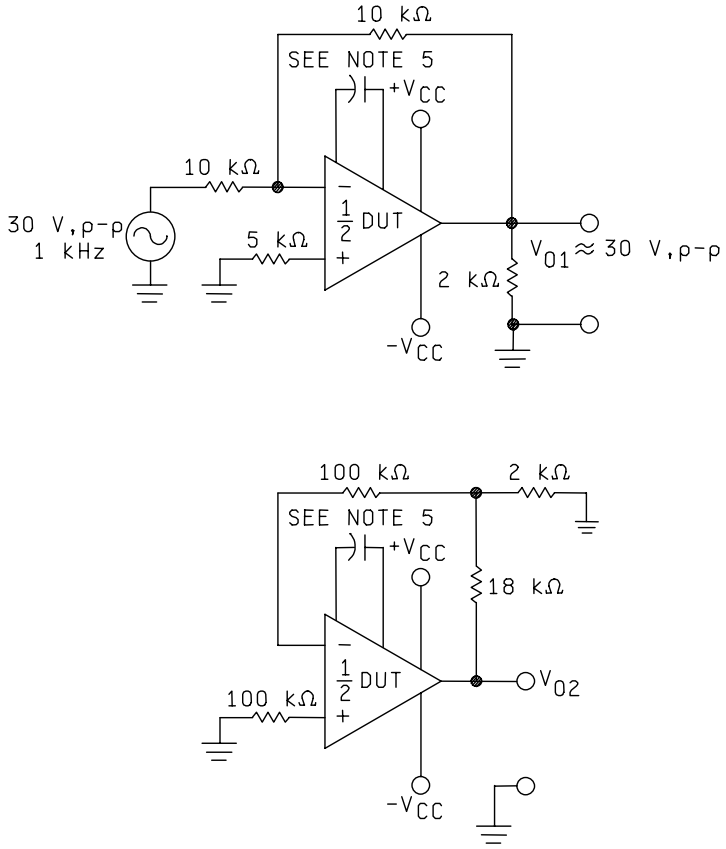
NOTES:

1. Idle half of dual amplifier shall be connected during test of other half.
2. All resistor tolerances are 1 percent, capacitor tolerances are 10 percent and $\pm V_{CC} = \pm 20$ V.
3. This compensation capacitor is used for device types 03, 04, 05, and 06.
4. For device types 01, 02, 03, 05, and 08, $R_L = 2$ kΩ; for device types 04 and 06, $R_L = 10$ kΩ.
5. Settling time is the interval from the beginning of the output response to the point where the output remains within the error band, in this case ±2 percent.
6. CF = 10 pF ±10 percent includes stray capacitance.
7. R1 may be added to the circuit. When R1 is added, its value shall be 10 kΩ. When using R1, the unity gain will increase to 2. To accommodate this change in gain, the pulse generator input shall be halved.
8. C1 may be added to the circuit. When added, it shall be within the range of 0 pF to 2 pF.
9. C_L capacitance specified includes stray, jig, and probe capacitance.

FIGURE 5. Transient response test circuit - Continued.

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Device types 02, 05, 06, and 08 only.

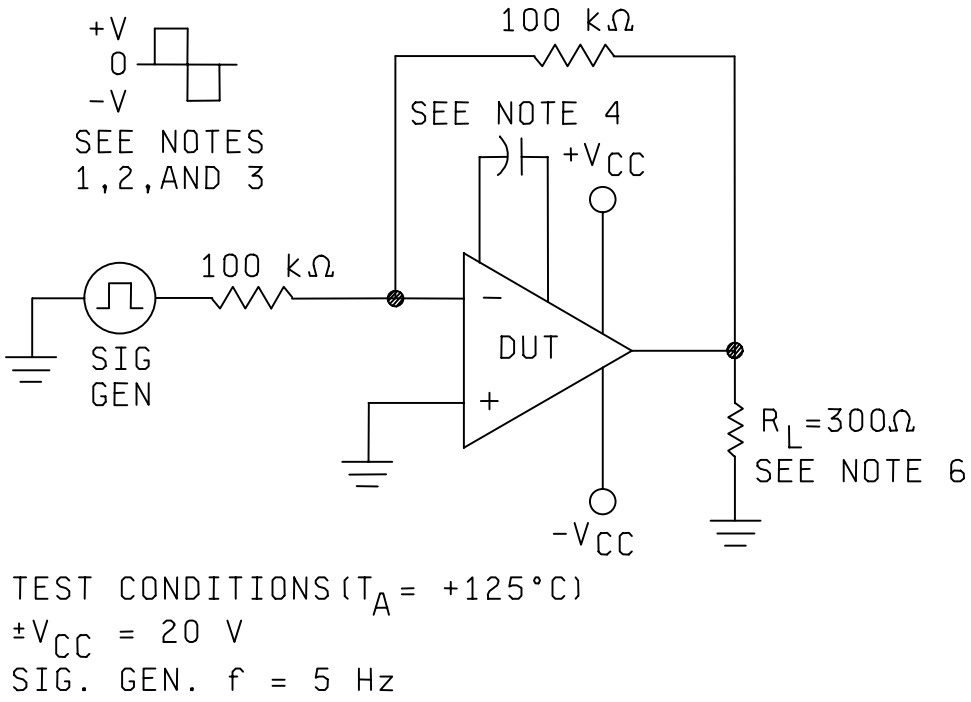


NOTES:

1. $\pm V_{CC}$ = 20 V.
2. Measure: V_{02} (volts, p-p) at 1 kHz to accuracy of 0.1 mV or better.
3. Channel separation (dB) referred to input of second channel = 20 log [V_{01} / (0.1 x V_{02})].
4. All resistor tolerances ≤ 1 percent.
5. A 30 pF compensation capacitor is required for device types 05 and 06.

FIGURE 6. Test circuit for channel separation.

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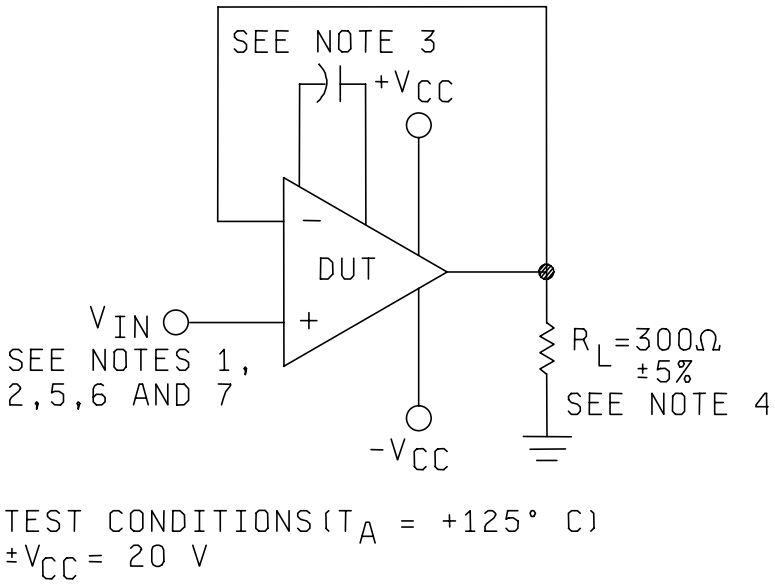


NOTES:

1. ±V = 3.0 V for device types 01, 03, 04, and 06.
2. ±V = 0.5 V for device types 02 and 08.
3. ±V = 1.0 V for device types 05 and 07.
4. A 30 pF compensation capacitor is required for types 03, 04, 05, and 06.
5. Both halves as above for dual amplifiers, types 02, 05, 06, and 08.
6. For device types 04 and 06, R_L = 1500 Ω.
7. All resistor tolerances are ±5 percent and shall not exceed its branded value due to use, heat or age.

FIGURE 7. Test circuit, burn-in and operating life test.

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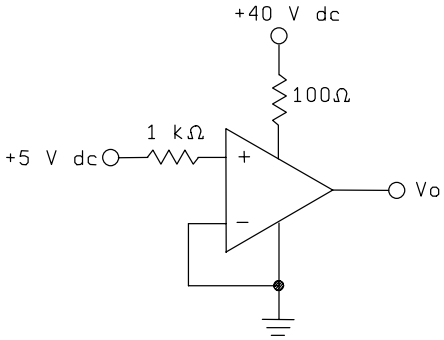
NOTES:

1. For device types 02 and 08, V_{IN} = 0.5 V dc, and shall be positive into one amplifier and negative into the other amplifier.
2. For device types 01 and 04, V_{IN} = 3.0 V dc. For device type 03, V_{IN} = +3.0 V or –3 V.
3. A 30 pF compensation capacitor is required for types 03, 04, 05, and 06.
4. For device types 04 and 06, R_L = 1500 Ω. The actual measured value of the resistor selected shall not exceed ±5 percent of its branded value due to use, heat, or age.
5. For device type 05, V_{IN} = 1.0 V dc, and shall be positive into one amplifier and negative in to the other amplifier.
6. For device type 06, V_{IN} = 3.0 V dc, and shall be positive into one amplifier and negative into the other amplifier.
7. For device type 07, V_{IN} = 1.0 V dc.

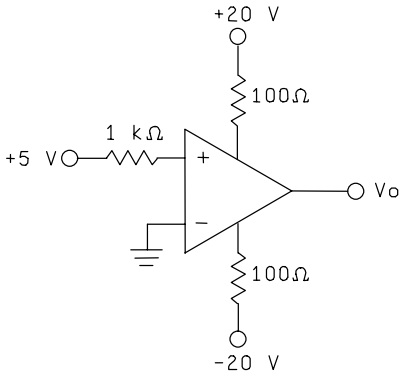
FIGURE 8. Test circuit, burn-in (steady state power and reverse bias) and operating life test.

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DEVICE TYPES 01,02,04,06,07, AND 08



DEVICE TYPES 03 AND 05

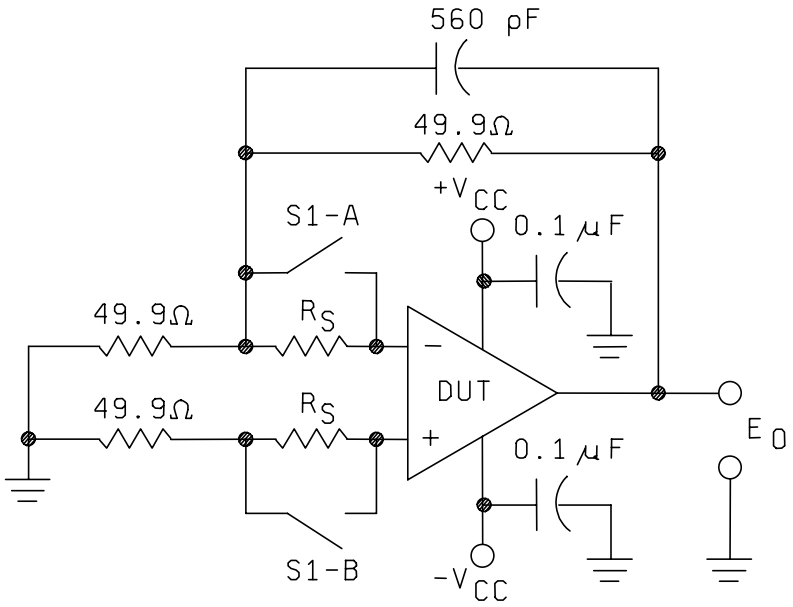


NOTES:

If accelerated, high temperature test conditions are used, the device manufacturer shall ensure that at least 85 percent of the applied voltage is dropped across the device during test at temperature. The device is not considered functional under accelerated test conditions. For dual devices both halves are connected as above. The actual measured value of the resistor selected shall not exceed ±5 percent of its branded value due to use, heat, or age.

FIGURE 9. Accelerated burn-in and life test circuit (or equivalent).

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Noise (Referred to input)	Symbol	S1	Measure		Measured equation	Parameter units
			Value	Units		
Broadband	N_1(BB)	Closed	E_0	mV rms	E_0 / 1000	µV rms
Popcorn	N_1(PC)	Open	E_0	mV pk	E_0 / 1000	µV pk

NOTES:

1. R_S = 20 kΩ for device types 01, 02, 07, and 08; R_S = 100 kΩ for device types 03, 04, 05, and 06.
2. E_0 is measured using an RMS voltmeter with a bandwidth of 10 Hz to 5 kHz and a peak detector simultaneously. Monitor the peak test for a minimum of 15 seconds. The loop bandwidth shall be at least 5 kHz.

FIGURE 10. Noise test circuit.

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TABLE III. Group A inspection.

Subgroup	Symbol	MIL-STD -883 method	Test no.	Conditions ±VCC = ±20 V dc, figure 4 unless otherwise specified	01, 02, 08 1/ Limits Min	01, 02, 08 1/ Limits Max	03, 05 1/ Limits Min	03, 05 1/ Limits Max	04, 06 1/ Limits Min	04, 06 1/ Limits Max	07 1/ Limits Min	07 1/ Limits Max	Unit
1 TA = +25°C	VIO	4001	1	VCM = -15 V 2/	-3.0	+3.0	-2.0	+2.0	-0.5	+0.5	-4.0	+4.0	mV
			2	VCM = +15 V 2/	"	"	"	"	"	"	"	"	"
			3	VCM = 0 V	"	"	"	"	"	"	"	"	"
			4	±VCC = ±5 V, VCM = 0 V	"	"	"	"	"	"	"	"	"
	IIO		5	VCM = -15 V 2/	-30	+30	-10	+10	-0.2	+0.2	-40	+40	nA
			6	VCM = +15 V 2/	"	"	"	"	"	"	"	"	"
			7	VCM = 0 V	"	"	"	"	"	"	"	"	"
			8	±VCC = ±5 V, VCM = 0 V	"	"	"	"	"	"	"	"	"
	+IIB		9	VCM = -15 V 2/	-0.1	110	-0.1	75	-0.1	+2.0	-0.1	250	nA
			10	VCM = +15 V 2/	"	"	"	"	"	"	"	"	"
			11	VCM = 0 V	"	"	"	"	"	"	"	"	"
			12	±VCC = ±5 V, VCM = 0 V	"	"	"	"	"	"	"	"	"
	-IIB		13	VCM = -15 V 2/	-0.1	110	-0.1	75	-0.1	+2.0	-0.1	250	nA
			14	VCM = +15 V 2/	"	"	"	"	"	"	"	"	"
			15	VCM = 0 V	"	"	"	"	"	"	"	"	"
			16	±VCC = ±5 V, VCM = 0 V	"	"	"	"	"	"	"	"	"
	+PSRR	4003	17	+VCC = 10 V, -VCC = -20 V	-50	+50	-50	+50	-16	+16	-100	+100	µV/V
	-PSRR		18	+VCC = 20 V, -VCC = -10 V	-50	+50	-50	+50	-16	+16	-100	+100	µV/V
	CMR		19	VCM = +15 V	80		80		96		80		dB
	VIO ADJ (+)		20	3/	+5		+4				+7		mV
	VIO ADJ (-)		21	3/		-5		-4				-7	mV

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TABLE III. Group A inspection - Continued.

Subgroup	Symbol	MIL-STD -883 method	Test no.	Conditions ±VCC = ±20 V dc, figure 4 unless otherwise specified	01, 02, 08 1/ Limits Min	01, 02, 08 1/ Limits Max	03, 05 1/ Limits Min	03, 05 1/ Limits Max	04, 06 1/ Limits Min	04, 06 1/ Limits Max	07 1/ Limits Min	07 1/ Limits Max	Unit
1 TA = +25°C	IOS (+) 4/	3011	22	±VCC = ±15 V, t ≤ 25 ms	-60		-60		-15		-65		mA
	IOS (-) 4/		23	±VCC = ±15 V, t ≤ 25 ms		+60		+60		+15		+65	mA
	ICC	4005	24	±VCC = ±15 V 5/		3.8		3.0		0.6		8	mA
2 TA = +125°C	VIO	4001	25	VCM = -15 V 2/	-4.0	+4.0	-3.0	+3.0	-1.0	+1.0	-6.0	+6.0	mV
			26	VCM = +15 V 2/	"	"	"	"	"	"	"	"	"
			27	VCM = 0 V	"	"	"	"	"	"	"	"	"
			28	±VCC = ±5 V, VCM = 0 V	"	"	"	"	"	"	"	"	"
	ΔVIO / ΔT 6/		29	ΔVIO / ΔT = [VIO (test 27) - VIO (test 3)] x 10	-15	+15	-15	+15	-5.0	+5.0	-50	+50	µV/°C
	IIO		30	VCM = -15 V 2/	-30	+30	-10	+10	-0.4	+0.4	-80	+80	nA
			31	VCM = +15 V 2/	"	"	"	"	"	"	"	"	"
			32	VCM = 0 V	"	"	"	"	"	"	"	"	"
			33	±VCC = ±5 V, VCM = 0 V	"	"	"	"	"	"	"	"	"
	ΔIIO / ΔT		34	ΔIIO / ΔT = [IIO (test 32) - IIO (test 7)] x 10	-200	+200	-100	+100	-2.5	+2.5	-1,000	+1,000	pA/°C
	+IIB		35	VCM = -15 V 2/	-0.1	110	-0.1	75	-1.0	+2.0	-0.1	250	nA
			36	VCM = +15 V 2/	"	"	"	"	"	"	"	"	"
			37	VCM = 0 V	"	"	"	"	"	"	"	"	"
			38	±VCC = ±5 V, VCM = 0 V	"	"	"	"	"	"	"	"	"
	-IIB		39	VCM = -15 V 2/	-0.1	110	-0.1	75	-1.0	+2.0	-0.1	250	nA
			40	VCM = +15 V 2/	"	"	"	"	"	"	"	"	"
			41	VCM = 0 V	"	"	"	"	"	"	"	"	"
			42	±VCC = ±5 V, VCM = 0 V	"	"	"	"	"	"	"	"	"

See footnotes at end of table III.

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TABLE III. Group A inspection - Continued.

Subgroup	Symbol	MIL-STD -883 method	Test no.	Conditions ±VCC = ±20 V dc, figure 4 unless otherwise specified	01, 02, 08 1/ Limits		03, 05 1/ Limits		04, 06 1/ Limits		07 1/ Limits		Unit
					Min	Max	Min	Max	Min	Max	Min	Max	
2 TA = +125°C	IOS (+) 4/	3011	43	±VCC = ±15 V, t ≤ 25 ms	-60		-60		-15		-65		mA
	IOS (-) 4/		44	±VCC = ±15 V, t ≤ 25 ms		+60		+60		+15		+65	mA
	ICC	4005	45	±VCC = ±15 V		3.4		2.5		0.6		7	mA
	+PSRR	4003	46	+VCC = 10 V, -VCC = -20 V	-100	+100	-100	+100	-16	+16	-150	+150	µV /V
	-PSRR		47	+VCC = 20 V, -VCC = -10 V	"	"	"	"	"	"	"	"	
	CMR	4003	48	VCM = ±15 V	80		80		96		80		dB
	VIO ADJ(+)		49	3/	+5		+4				+7		mV
	VIO ADJ(-)		50	3/		-5		-4				-7	
3 TA = -55°C	VIO	4001	51	VCM = -15 V 2/	-4.0	+4.0	-3.0	+3.0	-1.0	+1.0	-6.0	+6.0	mV
			52	VCM = +15 V 2/	"	"	"	"	"	"	"	"	"
			53	VCM = 0 V	"	"	"	"	"	"	"	"	"
			54	±VCC = ±5 V, VCM = 0 V	"	"	"	"	"	"	"	"	"
	ΔVIO / ΔT 6/		55	ΔVIO / ΔT = [VIO (test 3) - VIO (test 53)] x 12.5	-15	+15	-18	+18	-5.0	+5.0	-50	+50	µV /°C
	IIO		56	VCM = -15 V 2/	-70	+70	-20	+20	-0.4	+0.4	-80	+80	nA
			57	VCM = +15 V 2/	"	"	"	"	"	"	"	"	"
			58	VCM = 0 V	"	"	"	"	"	"	"	"	"
			59	±VCC = ±5 V, VCM = 0 V	"	"	"	"	"	"	"	"	"
	ΔIIO / ΔT 6/		60	ΔIIO / ΔT = [IIO (test 7) - IIO (test 58)] x 12.5	-500	+500	-200	+200	-2.5	+2.5	-1,000	+1,000	pA/°C

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TABLE III. Group A inspection - Continued.

Subgroup	Symbol	MIL-STD -883 method	Test no.	Conditions ±VCC = ±20 V dc, figure 4 unless otherwise specified	01, 02, 08 1/ Limits Min	01, 02, 08 1/ Limits Max	03, 05 1/ Limits Min	03, 05 1/ Limits Max	04, 06 1/ Limits Min	04, 06 1/ Limits Max	07 1/ Limits Min	07 1/ Limits Max	Unit
3 TA = -55°C	+IIB	4001	61	VCM = -15 V 2/	-0.1	265	-0.1	100	-0.1	3.0	-0.1	400	nA
			62	VCM = +15 V 2/	"	"	"	"	"	"	"	"	"
			63	VCM = 0 V	"	"	"	"	"	"	"	"	"
			64	±VCC = ±5 V, VCM = 0 V	"	"	"	"	"	"	"	"	"
	-IIB		65	VCM = -15 V 2/	-0.1	265	-0.1	100	-0.1	3.0	-0.1	400	nA
			66	VCM = +15 V 2/	"	"	"	"	"	"	"	"	"
			67	VCM = 0 V	"	"	"	"	"	"	"	"	"
			68	±VCC = ±5 V, VCM = 0 V	"	"	"	"	"	"	"	"	"
	IOS (+) 4/	3011	69	±VCC = ±15 V, t ≤ 25 ms	-60		-60		-15		-65		mA
	IOS (-) 4/		70	±VCC = ±15 V, t ≤ 25 ms		+60		+60		+15		80	mA
	ICC	4005	71	±VCC = ±15 V 5/		4.2		3.5		0.8		9	mA
	+PSRR	4003	72	+VCC = 10 V, -VCC = -20 V	-100	+100	-100	+100	-16	+16	-150	+150	µV/V
	-PSRR		73	+VCC = 20 V, -VCC = -10 V	-100	+100	-100	+100	-16	+16	-150	+150	µV/V
	CMR		74	VCM = ±15 V	80		80		96		80		dB
	VIO ADJ(+)		75	3/	+5		+4				+7		mV
	VIO ADJ(-)		76	3/		-5		-4				-7	mV
4 TA = +25°C	+VOP	4004	77	RL = 10 kΩ	+16		+16		+16		+17		V
	-VOP		78	RL = 10 kΩ		-16		-16		-16		-17	"
	+VOP		79	RL = 2 kΩ	+15		+15				+16		"
	-VOP		80	RL = 2 kΩ		-15		-15				-16	"

See footnotes at end of table III.

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TABLE III. Group A inspection - Continued.

Subgroup	Symbol	MIL-STD -883 method	Test no.	Conditions: ±VCC = ±20 V dc, figure 4 unless otherwise specified	01, 02, 08 1/ Limits Min	01, 02, 08 1/ Limits Max	03, 05 1/ Limits Min	03, 05 1/ Limits Max	04, 06 1/ Limits Min	04, 06 1/ Limits Max	07 1/ Limits Min	07 1/ Limits Max	Unit
4 TA = +25°C	+AVS	4004	81	RL = 2 kΩ, VOUT = +15 V	50		50				50		V/mV
	-AVS		82	RL = 2 kΩ, VOUT = -15 V	50		50				50		"
	+AVS		83	RL = 10 kΩ, VOUT = +15 V	50		50		80		50		"
	-AVS		84	RL = 10 kΩ, VOUT = -15 V	50		50		80		50		"
	AVS		85	RL = 2 kΩ, ±VCC = ±5 V, VOUT = ±2 V	10		10				10		"
	AVS		86	RL = 10 kΩ, ±VCC = ±5 V, VOUT = ±2 V	10		10		20		10		"
5 TA = +125°C	+VOP		87	RL = 10 kΩ	+16		+16		+16		+17		V
	-VOP		88	RL = 10 kΩ		-16		-16		-16		-17	"
	+VOP		89	RL = 2 kΩ	+15		+15				+16		"
	-VOP		90	RL = 2 kΩ		-15		-15				-16	"
	+AVS		91	RL = 2 kΩ, VOUT = +15 V	25		25				32		V/mV
	-AVS		92	RL = 2 kΩ, VOUT = -15 V	25		25				32		"
	+AVS		93	RL = 10 kΩ, VOUT = +15 V	25		25		40		32		"
	-AVS		94	RL = 10 kΩ, VOUT = -15 V	25		25		40		32		"
	AVS		95	RL = 2 kΩ, ±VCC = ±5 V, VOUT = ±2 V	10		10				10		"
	AVS		96	RL = 10 kΩ, ±VCC = ±5 V, VOUT = ±2 V	10		10		20		10		"
6 TA = -55°C	+VOP		97	RL = 10 kΩ	+16		+16		+16		+17		V
	-VOP		98	RL = 10 kΩ		-16		-16		-16		-17	"
	+VOP		99	RL = 2 kΩ	+15		+15				+16		"
	-VOP		100	RL = 2 kΩ		-15		-15				-16	"

See footnotes at end of table III.

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TABLE III. Group A inspection - Continued.

Subgroup	Symbol	MIL-STD -883 method	Test no.	Conditions ±VCC = ±20 V dc, figure 4 unless otherwise specified	01, 02, 08 1/ Limits		03, 05 1/ Limits		04, 06 1/ Limits		07 1/ Limits		Unit
					Min	Max	Min	Max	Min	Max	Min	Max	
6 TA = -55°C	+AVS		101	RL = 2 kΩ, VOUT = +15 V	25		25				32		V/mV
	-AVS		102	RL = 2 kΩ, VOUT = -15 V	25		25				32		"
	+AVS		103	RL = 10 kΩ, VOUT = +15 V	25		25		40		32		"
	-AVS		104	RL = 10 kΩ, VOUT = -15 V	25		25		40		32		"
	AVS		105	RL = 2 kΩ, ±VCC = ±5 V, VOUT = ±2 V	10		10				10		"
	AVS		106	RL = 10 kΩ, ±VCC = ±5 V, VOUT = ±2 V	10		10		20		10		"
7 TA = +25°C	TR(tr)		107	Figure 5		800		800		1,000		40	ns
	TR(OS)		108	Figure 5		25		25		50		50	%
	SR(+)	4002	109	Figure 5, AV = 1, VIN = -5 V to +5 V	0.3		0.3		0.05		40		V/µs
	SR(-)		110	Figure 5, AV = 1, VIN = +5 V to -5 V	0.3		0.3		0.05		40		V/µs
	CS 7/		111	Figure 6	80		80		80				dB
	NI(BB)		112	BW = 5 kHz, figure 10		15		15		15		25	µV rms
	NI(PC)		113	BW = 5 kHz, figure 10		40		80		40		80	µV pk
8A TA = +125°C	TR(tr)		114	TA = +125°C, figure 5		800		800		1,000		40	ns
	TR(OS)		115	TA = +125°C, figure 5		25		25		50		50	%
	SR(+)	4002	116	TA = +125°C, figure 5, AV = 1, VIN = -5 V to +5 V	0.3		0.3		0.05		30		V/µs
	SR(-)		117	TA = +125°C, figure 5, AV = 1, VIN = +5 V to -5 V	0.3		0.3		0.05		30		V/µs
8B TA = -55°C	TR(tr)		118	TA = -55°C, figure 5		800		800		1,000		40	ns
	TR(OS)		119	TA = -55°C, figure 5		25		25		50		50	%
	SR(+)	4002	120	TA = -55°C, figure 5, AV = 1, VIN = -5 V to +5 V	0.3		0.2		0.05		40		V/µs
	SR(-)		121	TA = -55°C, figure 5, AV = 1, VIN = +5 V to -5 V	0.3		0.2		0.05		40		V/µs

See footnotes at end of table III.

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TABLE III. Group A inspection - Continued.

Subgroup	Symbol	MIL-STD -883 method	Test no.	Conditions $\pm V_{CC}$ = ±20 V dc, figure 4 unless otherwise specified	01, 02, 08 1/ Limits		03, 05 1/ Limits		04, 06 1/ Limits		07 1/ Limits		Unit
					Min	Max	Min	Max	Min	Max	Min	Max	
12	$t_{S(+)}$	4002	122	T_A = +25°C, figure 5								800	ns
T_A =+25°C	$t_{S(-)}$		123	T_A = +25°C, figure 5								800	ns
13A	$t_{S(+)}$		124	T_A = +125°C, figure 5								1,200	ns
T_A =+125°C	$t_{S(-)}$		125	T_A = +125°C, figure 5								1,200	ns
13B	$t_{S(+)}$		126	T_A = -55°C, figure 5								1,200	ns
T_A =-55°C	$t_{S(-)}$		127	T_A = -55°C, figure 5								1,200	ns

1/ For devices marked with the "Q" certification mark, the parameters listed herein may be guaranteed if not tested to the limits specified herein in accordance with the manufacturer's QM plan. Limits apply to both halves of dual devices (02, 05, 06, and 08) independently, and slew rate limit for device 08 is 0.4 V/µs at all temperatures (tests 109, 110, 116, 117, 120, 121).

2/ V_{CM} is achieved by algebraically subtracting the common mode voltage from each supply and algebraically adding the common mode voltage to V (example, for V_{CM} = -15 V, +V_{CC} = +35 V, -V_{CC} = -5 V, V = -15 V).

3/ V_{IO} (ADJ) is not performed on device type 02, case I only, or on device types 04, 06, and 08 all case types.

4/ Due to the significant power dissipation and associated device heating, these tests shall always be the last tests performed in any given sequence, followed by operational verification (example, such tests at V_{OPP}, AVS, TR, SR).

5/ Limit shown applied to single devices (01, 03, and 04) only. The maximum quiescent I_{CC} for dual devices (02, 05, 06, and 08) is twice that shown for single devices.

6/ Tests 29, 34, 55, and 60 which require a read and record measurement plus a calculation, may be omitted except when subgroups 2 and 3 are being accomplished for group A sampling inspection and group C and D endpoint measurements.

7/ Applies to device types 02, 05, and 06 only.

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TABLE IV. Group C end point electrical parameters.

(T_A = +25°C, V_{CC} = ±20 V, V_{CM} = 0 V)

Table III test no.	Test	01, 02, 08				03, 05				Unit
		Limit		Delta		Limit		Delta		
		Min	Max	Min	Max	Min	Max	Min	Max	
3	V_{IO}	-3.0	+3.0	-0.5	+0.5	-2.0	+2.0	-0.5	+0.5	mV
11	$+I_{IB}$	+1.0	+110	-12	+12	+1.0	+75	-7.5	+7.5	nA
15	$-I_{IB}$	+1.0	+110	-12	+12	+1.0	+75	-7.5	+7.5	nA

Table III test no.	Test	04, 06				07				Unit
		Limit		Delta		Limit		Delta		
		Min	Max	Min	Max	Min	Max	Min	Max	
3	V_{IO}	-0.5	+0.5	-0.25	+0.25	-4.0	+4.0	-1.0	+1.0	mV
11	$+I_{IB}$	-0.1	+2.0	-0.5	+0.5	+1.0	250	-25	+25	nA
15	$-I_{IB}$	-0.1	+2.0	-0.5	+0.5	+1.0	250	-25	+25	nA

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4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510 and as specified herein. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883, and herein for groups A, B, C, and inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Subgroups 9, 10, and 11 shall be omitted.

b. Tests shall be as specified in table II herein.

c. Subgroups 12 and 13 (for device type 07 only) shall be added to table I of method 5005 of MIL-STD-883 for class S only. The class S sample size series for subgroup 12 shall be 5 and for subgroup 13 the class S sample size series shall be 7.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883 and as follows:

a. Electrical parameters shall be as specified in table II herein.

b. Steady state life test for class S devices shall be in accordance with table IIa of method 5005 of MIL-STD-883, using the circuit shown on figure 9. If alternate burn-in conditions are used, the circuit shown on either figure 7 or figure 8 shall be used.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III, method 5005 of MIL-STD-883 and as follows:

a. End point electrical parameters shall be as specified in table II herein. Delta limits shall apply only to subgroup 1 of group C inspection for class B devices.

b. Life test for class B (method 1005 of MIL-STD-883). Test condition D using the circuit shown on figure 7 or test condition C using the circuit shown on figure 8, or test condition F using the circuit shown on figure 9.

c. Subgroups 3, 4, and 5 shall be added to group C inspection and shall consist of subgroups 8, 12, and 13 respectively as specified in table III herein. The sample size series for subgroup 12 shall be 5, and subgroup 13 shall be 7 for class B devices (see MIL-PRF-38535, Appendix D).

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End point electrical parameters shall be as specified in table II herein.

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4.5 Methods of inspection. The methods of inspection shall be specified in the appropriate tables. Electrical test circuits as prescribed herein or in the referenced test methods of MIL-STD-883 shall be acceptable. Other test circuits shall require the approval of the qualifying activity.

4.5.1 Voltage and current. All voltage values given, except the input offset voltage (or differential voltage) are referenced to the external zero reference level of the supply voltage. Currents given are conventional current and positive when flowing into the referenced terminal.

4.5.2 Burn-in and life test cooldown procedure. When devices are measured at +25°C following application of the steady state life or burn-in condition, they shall be cooled to within 10°C of their power stable condition at room temperature prior to removal of the bias.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for Government microcircuit applications (original equipment) and logistic purposes.

6.2 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List QML-38535 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DSCC-VQ, 3990 E. Broad Street, Columbus, Ohio 43213-1199.

6.3 Ordering data. The contract or purchase order should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to acquiring activity in addition to notification of the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of MIL-STD-883, method 5003), corrective action and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special lead lengths or lead forming, if applicable. These requirements shall not affect the part number.

h. Requirements for "JAN" marking.

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6.4 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510 and MIL-STD-1331.

6.5 Logistic support. Lead materials and finishes (see 3.4) are interchangeable. Unless otherwise specified, microcircuits acquired to Government logistic support will be acquired to device class B (see 1.2.2), and lead material and finish C (see 3.4). Longer length leads and lead forming shall not affect the part number.

6.6 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	741A
02	747A (with common $+V_{CC}$)
03	LM101A
04	LM108A
05	LH2101A
06	LH2108A
07	LM118
08	1558

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - NW
Air Force - 11
NASA - NA
DLA - CC

Review activities:
Army - AR, MI, SM
Navy - AS, CG, OS, SH, TD
Air Force - 19, 71, 99

Preparing activity:
DLA - CC

Project 5962-1915

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7, and send to preparing activity.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-M-38510/101H	2. DOCUMENT DATE *(YYYYMMDD)*

3. DOCUMENT TITLE

MICROCIRCUITS, LINEAR, OPERATIONAL AMPLIFIERS, MONOLITHIC SILICON, PART NUMBER M38510/10101 THROUGH M38510/10108

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) DSN *(If applicable)*	7. DATE SUBMITTED *(YYYYMMDD)*

8. PREPARING ACTIVITY

a. NAME Rick Officer	b. TELEPHONE *(Include Area Code* (1) Commercial 614-692-0518	(2) DSN 850-0518
c. ADDRESS *(Include Zip Code)* DSCC-VAS 3990 East Broad Street Columbus, Ohio 43216-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman Road, Suite 2533 Fort Belvoir, Virginia 22060-6221 Telephone (703)767-6888 DSN 427-6888	

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WHS/DIOR, Feb 99

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